Complete Lagrangian of the PSP Model

E.V. Chernoknizhny
Independent Researcher
ORCID: 0009-0007-2558-9172
chernocnijniy@yandex.ru

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Lagrangian of the Quantized PSP Field

The PSP model is described by a unified Lagrangian that combines the quantized structure of space, gravity, matter flows, and structure formation.

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\begin{aligned}
\mathcal{L}_{\text{PSP}} = & \underbrace{\frac{1}{2}(\partial_\mu \psi \partial^\mu \psi - m^2\psi^2) - \frac{\lambda}{4}\psi^4 + \frac{1}{2}(\partial_\mu M \partial^\mu M + 1)^2}_{\text{Quantized Field}} \\
& + \underbrace{\frac{1}{16\pi G}R - \frac{G M_{\text{tor}} M_Q}{R_{\text{tor}}} - \frac{1}{2}\omega_{\text{tor}}^2 R_{\text{tor}}^2}_{\text{Gravity and Geometry}} \\
& + \underbrace{\frac{1}{2}\partial_\mu J^\mu_{\text{jet}} \partial^\mu J^\mu_{\text{jet}} - \frac{1}{2}m^2_{\text{jet}} J^2_{\text{jet}} + g_{\text{jet}} J^\mu_{\text{jet}} \bar{\psi}_p \psi_n}_{\text{Jet Flow}} \\
& + \underbrace{\frac{1}{2}\partial_\mu J^\mu_{\text{tor}} \partial^\mu J^\mu_{\text{tor}} - \frac{1}{2}m^2_{\text{tor}} J^2_{\text{tor}} + g_{\text{tor}} J^\mu_{\text{tor}} \bar{\psi}_p \psi_n}_{\text{Torus Flow}} \\
& + \underbrace{-\frac{1}{2}\sigma_{\text{shell}}(\nabla M \cdot \Phi_M)^2 - \frac{1}{2}\eta_{\text{turb}}(\partial_\mu \mathcal{F})^2 - \frac{1}{4}\kappa(\mathcal{F}^2 - \mathcal{F}_0^2)^2}_{\text{Structure Formation}} \\
& + \underbrace{-\frac{1}{2}\omega_{\text{tor}}^2 R_{\text{tor}}^2 + \frac{1}{2}\partial_\mu R_{\text{shell}} \partial^\mu R_{\text{shell}} - \frac{1}{2}\gamma \dot{M}_Q^2}_{\text{Sources and Dissipation}}
\end{aligned}
}
$$

Physical Interpretation of Terms

1. Quantized Field $\mathcal{L}_{\text{field}}$

$$
\mathcal{L}_{\text{field}} = \frac{1}{2}(\partial_\mu \psi \partial^\mu \psi - m^2\psi^2) - \frac{\lambda}{4}\psi^4 + \frac{1}{2}(\partial_\mu M \partial^\mu M + 1)^2
$$

- ψ — protomatter field (quantum nodes).
- $M \in [0, 1]$ — cycle phase.
- $-\frac{\lambda}{4}\psi^4$ — describes the phase transition from chaos ($M < 0$) to cycle.

- $\frac{1}{2}(\partial_\mu M \partial^\mu M + 1)^2$ — sets phase discreteness.

2. Gravity $\mathcal{L}_{\mathbf{grav}}$

$$\mathcal{L}_{\text{grav}} = \frac{1}{16\pi G} R - \frac{G M_{\text{tor}} M_Q}{R_{\text{tor}}} - \frac{1}{2}\omega_{\text{tor}}^2 R_{\text{tor}}^2$$

- R — scalar curvature of the field.
- M_{tor}, M_Q — masses of the torus and quasar.
- ω_{tor} — torus rotation frequency.

3. Flows $\mathcal{L}_{\mathbf{flow}}$

$$\mathcal{L}_{\text{flow}} = \frac{1}{2}\partial_\mu J_{\text{jet}}^\mu \partial^\mu J_{\text{jet}}^\mu - \frac{1}{2} m_{\text{jet}}^2 J_{\text{jet}}^2 + g_{\text{jet}} J_{\text{jet}}^\mu \bar{\psi}_p \psi_n + \frac{1}{2}\partial_\mu J_{\text{tor}}^\mu \partial^\mu J_{\text{tor}}^\mu - \frac{1}{2} m_{\text{tor}}^2 J_{\text{tor}}^2 + g_{\text{tor}} J_{\text{tor}}^\mu \bar{\psi}_p \psi_n$$

- J_{jet}^μ — quasar jet flow.
- J_{tor}^μ — torus rotation flow.
- $g_{\text{jet}}, g_{\text{tor}}$ — flow-matter coupling constants.
- ψ_p, ψ_n — proton and neutron fields.

4. Structures $\mathcal{L}_{\mathbf{struct}}$

$$\mathcal{L}_{\text{struct}} = -\frac{1}{2}\sigma_{\text{shell}}(\nabla M \cdot \Phi_M)^2 - \frac{1}{2}\eta_{\text{turb}}(\partial_\mu \mathcal{F})^2 - \frac{1}{4}\kappa(\mathcal{F}^2 - \mathcal{F}_0^2)^2$$

- σ_{shell} — field surface tension (forms spiral arms).
- η_{turb} — field turbulence coefficient.
- κ — field condensation constant (star formation).
- $\mathcal{F} = \mathcal{E}_0 \cdot (R_{\text{shell}}/R_{\text{shell}}(M_0))^{-0.581} \cdot (\nabla M \cdot \Phi_M) \cdot \sum_i q_i \delta(x - x_i)\delta(M - M_i)$ — quantized field.

5. Sources and Dissipation $\mathcal{L}_{\mathbf{source}}$

$$\mathcal{L}_{\text{source}} = -\frac{1}{2}\omega_{\text{tor}}^2 R_{\text{tor}}^2 + \frac{1}{2}\partial_\mu R_{\text{shell}} \partial^\mu R_{\text{shell}} - \frac{1}{2}\gamma \dot{M}_Q^2$$

- R_{shell} — shell radius.
- γ — dissipation coefficient (creates the global rhythm $T_0 = 16.35$ days).
- $\dot{M}_Q$ — quasar accretion rate.

Consequences of the Lagrangian

From $\mathcal{L}_{\text{PSP}}$ all key effects of the model follow:

Effect	**Lagrangian Term**
Phase quantization $M_n = n/N$	$\frac{1}{2}(\partial_\mu M \partial^\mu M + 1)^2$
Cycle birth	$-\frac{\lambda}{4}\psi^4$
Gravity	$\frac{1}{16\pi G}R - \frac{GM_{\text{tor}}M_Q}{R_{\text{tor}}}$
Spiral arms	$-\frac{1}{2}\sigma_{\text{shell}}(\nabla M \cdot \Phi_M)^2$
Star formation	$-\frac{1}{4}\kappa(\mathcal{F}^2 - \mathcal{F}_0^2)^2$
Global rhythm $T_0 = 16.35$ days	$-\frac{1}{2}\gamma \dot{M}_Q^2$

Table 1: Connection between Lagrangian terms and observable effects

The Law of Quantized PSP Field

The field in the PSP model is defined as:

$$\mathcal{F}(x, M) = \mathcal{E}_0 \cdot \left(\frac{R_{\text{shell}}(M)}{R_{\text{shell}}(M_0)} \right)^{-0.581} \cdot (\nabla M \cdot \Phi_M) \cdot \sum_i q_i \delta(x - x_i)\delta(M - M_i)$$

where:

- $\mathcal{E}_0$ — field energy at the calibration point ($M_0 = 0.29$).
- $\alpha = 0.581$ — Fuger law exponent.
- $\hat{n} = \nabla M \cdot \Phi_M$ — event vector.
- q_i — quanta of connections between field nodes.

Conclusion

The Lagrangian $\mathcal{L}_{\text{PSP}}$ describes a unified quantized medium in which all structures — from quantum fluctuations to spiral arms of galaxies and stars — are consequences of a single fundamental field. The model does not require dark matter or dark energy as separate entities.